

HAFIZ MUHAMMAD USAMA SALEEM

Data Scientist | AI Engineer | Generative AI and Agentic AI

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [RiVara](#)

PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 3+ years of experience delivering production grade Machine Learning, Generative AI, LLM, Retrieval Augmented Generation, and Agentic AI solutions across commercial and operational domains. Experienced across the full lifecycle including data preprocessing, feature engineering, model development, fine tuning, evaluation, retrieval optimization, API development, containerization, cloud deployment, and monitoring of AI systems in production. Proficient in Python, SQL, FastAPI, PyTorch, TensorFlow, LangChain, LangGraph, CrewAI, vector databases, Docker, AWS, GCP, Git, and CI/CD, with strong ability to translate complex business questions into scalable data science solutions and communicate with non technical stakeholders. Currently completing an MSc in Data Science, expected December 2026, researching reasoning consistency in LLM based agents.

TECHNICAL SKILLS

- Programming and Data: Python, SQL, Pandas, NumPy, Scikit learn, Data Preprocessing, Feature Engineering, Exploratory Data Analysis, ETL
- Machine Learning and Deep Learning: Machine Learning, Deep Learning, Supervised Learning, Unsupervised Learning, Transformers, PyTorch, TensorFlow, Keras, Model Evaluation, Experimentation and A/B Testing
- Generative AI and LLMs: Generative AI, LLMs, Prompt Engineering, Context Engineering, LLM Integration, LLM Fine Tuning, RAG, Embeddings, Information Extraction
- Agentic AI: AI Agents, Agentic AI, Multi Agent Systems, Tool Use, LangGraph, LangChain, CrewAI, SmolAgent, Semantic Kernel concepts
- AI Engineering: Production AI Applications, FastAPI, REST APIs, Model Deployment, AI Evaluation and Monitoring, Scalable AI Application Development, Reusable Libraries and Frameworks
- Databases: PostgreSQL, MySQL, Vector Databases, OpenSearch, FAISS
- Cloud and DevOps: AWS, Google Cloud Platform, Vertex AI, Docker, Git, GitHub Actions, CI/CD, MLOps Workflows
- NLP and Voice AI: NLP, Text Classification, Summarization, Whisper, Speech to Text, Text to Speech, Twilio
- Tools: Hugging Face, Jupyter Notebook, Streamlit, Cursor, GitHub

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, PixelEdge Solutions (Pvt.) Limited *Oct 2023 to Apr 2026*

- Designed and deployed production-oriented AI solutions using Python, FastAPI, LLMs, Machine Learning, Docker, AWS, and CI/CD, aligning technical delivery with measurable business outcomes.
- Built an AI Voice Assistant integrating LLMs, Speech to Text, Text to Speech, Whisper, and Twilio to automate customer interactions and reduce manual support effort.
- Architected Retrieval Augmented Generation pipelines using LangChain, embeddings, vector search, and OpenSearch for document intelligence and knowledge retrieval.
- Fine-tuned and integrated open-source LLMs for domain specific applications, improving task performance while reducing inference and licensing costs.
- Developed Agentic AI and multi agent workflows using LangGraph, CrewAI, and SmolAgent for research, information extraction, reasoning, and structured report generation.
- Developed reusable, production grade FastAPI REST APIs for AI and ML services and integrated retrieval and generation capabilities into business applications.
- Built Machine Learning workflows for financial data processing, feature engineering, KPI forecasting, and automated reporting using Python, Pandas, NumPy, and Scikit learn.
- Containerized AI services with Docker and supported cloud deployment, monitoring, and production delivery using AWS, Git, GitHub Actions, and CI/CD, following MLOps best practices.

Trainee Data Scientist, Techma Zone *Jan 2023 to Oct 2023*

- Performed data preprocessing, data cleaning, feature engineering, and exploratory data analysis on real world datasets to support business decision making.
- Developed and evaluated Machine Learning models using Python, Scikit learn, Pandas, and NumPy.
- Created data visualizations using Matplotlib, Seaborn, and Plotly and communicated analytical findings to cross functional project teams.
- Collaborated on cross functional data science projects involving model development, experimentation, and evaluation.

SELECTED AI AND MACHINE LEARNING PROJECTS

RiVara | AI Powered SaaS Platform | [Live Platform](#)

- Designed and developed an AI powered SaaS platform focused on business workflow automation using LLMs, AI agents, knowledge sources, conversations, and configurable AI workflows.
- Built modular application architecture with backend services, database integration, organization management, knowledge sources, and conversational workflows.

AI Data Analyst Agent | CSV and Excel Demo | [Demo Video](#)

- Built an AI powered data analysis assistant that allows users to upload CSV or Excel files and ask questions in natural language.
- Designed the system to generate and execute Pandas operations on uploaded data for reliable analytical results, supporting follow up questions and large result sets.

Mini AI Concierge | Guest Support Chatbot | [GitHub](#) | [Demo Video](#)

- Built an LLM powered guest support chatbot using RAG, FastAPI, FAISS, and Gemini to retrieve information from property documents and provide automated support.
- Implemented document retrieval and fallback escalation logic for handling unsupported or uncertain queries.

Agentic Travel Companion | Multi Agent Trip Planner | [GitHub](#) | [Demo Video](#)

- Developed a Streamlit application using multiple specialized AI agents and CrewAI for destination selection, budgeting, and itinerary generation.

Stock Price Prediction Model | [GitHub](#)

- Developed a Logistic Regression model to predict stock price movement using financial indicators, exploratory data analysis, and feature selection.

EDUCATION

MSc in Data Science

FAST National University of Computer and Emerging Sciences (FAST NUCES), Expected December 2026

Thesis: Evaluating Reasoning Consistency in LLM Based Agents. Research focuses on the reliability and consistency of decisions produced by LLM based agents using reasoning consistency, decision stability, and trajectory divergence measures.

Postgraduate Diploma in Data Science and Artificial Intelligence

NED University, 2024

Bachelor of Engineering in Electrical Engineering

NED University, 2020

CERTIFICATIONS

- Generative AI with Large Language Models, Coursera
- Mathematics for Machine Learning and Data Science Specialization, Coursera
- Complete Machine Learning and Data Science Bootcamp, Udemy
- 100 Days of Code: The Complete Python Pro Bootcamp, Udemy

CORE COMPETENCIES

Problem Solving | Analytical Thinking | Stakeholder Communication | Adaptability | Time Management | Cross Functional Collaboration

Languages: English (Professional Working Proficiency), Urdu (Native)